

# Erdős Problem #1093

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| EP-1093 | L1 - Tractable | Partiellement résolu | Number Theory |
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## Énoncé original (anglais, reproduction mot pour mot)

For  $n \geq 2k$  we define the deficiency of  $\binom{n}{k}$  as follows. If  $\binom{n}{k}$  is divisible by a prime  $p \leq k$  then the deficiency is undefined. Otherwise, the deficiency is the number of  $0 \leq i < k$  such that  $n-i$  is  $k$ -smooth, that is, divisible only by primes  $\leq k$ .

Are there infinitely many binomial coefficients with deficiency 1? Are there only finitely many with deficiency  $> 1$ ?

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